

Callum Parker

LECTURER IN INTERACTION DESIGN

Sydney, Australia

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Experience

Lecturer in Interaction Design

DESIGN LAB, THE UNIVERSITY OF SYDNEY

[Sydney, Australia](#)

January 2019 - Present

- Continuing Research, Teaching and Service role (split 40/40/20) since 2019.

Program Director, Bachelor of Design (Interaction Design) & Bachelor of Design Computing

DESIGN LAB, THE UNIVERSITY OF SYDNEY

[Sydney, Australia](#)

December 2024 - Present

- Overseeing and managing the entire degree programs. This role involves student-facing tasks, such as responding to queries about aspects like progression and enrollment. It also involves managing the units within the degree and the coordinators who run them, ensuring units are run smoothly and consistently across the program.

Lead Coordinator (Director) of the Urban Interfaces Lab

THE UNIVERSITY OF SYDNEY

[Sydney, Australia](#)

January 2022 - Present

- Leading an interdisciplinary research group of 9 academics and 10 research students, exploring emerging technologies in cities.

Research Fellow

INNOVATION IN APPLIED DESIGN (IAD) LAB, THE UNIVERSITY OF SYDNEY

[Sydney, Australia](#)

May 2017 - June 2019

- Worked on a research project in collaboration with LendLease, a multinational construction firm.
- Researched how augmented reality can be utilised to improve the safety and performance of workers prefabricating building components.
- Developed prototype augmented reality apps for head mounted displays and smartphones.
- Conducted user studies in the lab and in-situ, to evaluate the developed apps and their integration into factory and construction site workflows.

Simulation Developer

SCHOOL OF PSYCHOLOGY AND THE AUSTRALIAN DEPARTMENT OF DEFENCE

[Sydney, Australia](#)

2016

- Working with the School of Psychology at the University of Sydney, and the Australian Department of Defence, I developed a virtual reality simulation for user studies which could automatically collect statistics about a participant's session and feed them into a Bayesian network to make automated decisions about the user.

Researcher/Developer

SCHOOL OF ARCHITECTURE, DESIGN AND PLANNING, THE UNIVERSITY OF SYDNEY

[Sydney, Australia](#)

2015 - 2016

- Worked in a team to develop a digital popup for community engagement - consisting of: a social drawing robot, selfie booth, and a transit touchscreen system.
- Assisted with the study of this popup, where I observed people walking past and interacting with the booth and conducted interviews.

Augmented Reality Developer

FREELANCE FOR AN AURECON AND AUSGRID COMMUNITY CONSULTATION PROJECT

[Sydney, Australia](#)

2015

- Developed an outdoor augmented reality game using Unity, Google Cardboard, and the Vuforia augmented reality library.

Education

Graduate Certificate in Educational Studies (Higher Education)

THE UNIVERSITY OF SYDNEY

[Sydney, Australia](#)

2022

- In this year-long course, I developed my knowledge of scholarly, research-led approaches to teaching while engaging in activities and tasks aimed at cultivating critical reflection about higher education.

Doctor of Philosophy (Human-Computer Interaction)

THE UNIVERSITY OF SYDNEY

[Sydney, Australia](#)

2014 - 2019

- Thesis Title: "Augmenting Space: Design approaches for making public interactive displays relevant in hyperconnected societies".
- PhD conferred in April 2019.
- Thesis accepted without revisions.

Bachelor of Computing

UNIVERSITY OF TASMANIA

[Launceston, Australia](#)

2011 - 2013

- Majored in Human Interface Technology.

University Service

Design @ Dusk

ORGANISER

Sydney, Australia

March 2016 - May 2020

- Organised speakers, schedule and venue.
- Sent out reminders to staff and students to attend this event.

Bachelor of Design Computing Honours Program

COORDINATOR

Sydney, Australia

February 2019 - December 2020

- Creation of a formalised unit of study outline.
- Re-developed the dissertation marking criteria.
- Managed Honours students and dissertation examiners.

Bachelor of Design Computing Program

ACTING PROGRAM DIRECTOR

Sydney, Australia

June 2019 - August 2020

- Responded to student queries.
- Handled course transfers, credit, marking.
- Managed unit coordination staff.

Teaching Experience

In my current role as a Lecturer in Interaction Design, 40% of my time is reserved for teaching duties, including giving lectures, unit coordination, and in-class teaching. I teach three units per year, split across two 13-week semesters.

SEMESTER 1

Designing for Virtual Reality (DECO3551/DECO3009)

COORDINATOR/LECTURER

2017 - Present

- This is an elective unit for undergraduate Bachelor of Design (Interaction Design) students. My role of this unit is managing teaching staff, creating Unity and virtual reality (VR) app development content (regularly updating based on constant changes in the industry), and delivering lectures on history, design practices, and development of VR.

SEMESTER 2

Interactive Product Design Studio (DECO3200/DECO4200)

COORDINATOR/LECTURER

2020 - Present

- This is a capstone project unit for undergraduate Bachelor of Design (Interaction Design) students. My role for this unit is to manage staff, respond to student queries and create content to help support students completing their advanced design capstone project.

Lab 2 (IDEA9201)

COORDINATOR/LECTURER

2021 - 2023

- This is a core unit for the Master of Interaction Design and Electronic Arts (MIDEA) students. My role of this unit is managing teaching staff, creating Unity and mixed reality (MR) app development content (regularly updating based on constant changes in the industry), and delivering lectures on history, design practices, and development of MR.

PAST UNITS

Design programming (DECO1012)

TUTOR

2016 - 2019

- I taught first-year students Processing (Java) and P5js (JavaScript).
- I also delivered lecture content covering key concepts of coding (best practices, syntax, and algorithms).

3D modelling (DECO1008), Design for Social Media (DECO2010), Information Visualisation (DECO3100)

TUTOR

2014

- Across these units I taught students:
 - > Rhino and Grasshopper for 3D modelling.
 - > Design methods for effective social media campaigns.
 - > Creating visualisations in the following programming languages: R, Python, and JavaScript.

Biology (BIOL1008)

TUTOR

2016

- I taught video editing and Java programming to biology students.

Supervision

CURRENT

Jimin Lee PHD CANDIDATE	<i>Lead Supervisor</i> 2025 - Present
Samar Albujabir PHD CANDIDATE	<i>Auxiliary Supervisor</i> 2025 - Present
Dylan Cheng PHD CANDIDATE	<i>Auxiliary Supervisor</i> 2022 - Present
Tianrui Liu PHD CANDIDATE	<i>Auxiliary Supervisor</i> 2024 - Present

COMPLETED

Thi Minh Tram Tran PHD CANDIDATE	<i>Lead Supervisor</i> 2020 - 2024
Adrian Ramos HONOURS STUDENT	<i>Lead Supervisor</i> 2024
Nicole Moran HONOURS STUDENT	<i>Lead Supervisor</i> 2023
Natsumi Wang HONOURS STUDENT	<i>Lead Supervisor</i> 2023
Rachel Montgomery HONOURS IN DESIGN COMPUTING	<i>Lead supervisor</i> 2021
Tina Yao HONOURS IN DESIGN COMPUTING	<i>Lead supervisor</i> 2019
Thi Minh Tram Tran MASTER OF INTERACTION DESIGN AND ELECTRONIC ARTS (MIDEA)	<i>Lead supervisor</i> 2019
Tony “Teddy” Wang MASTER OF INTERACTION DESIGN AND ELECTRONIC ARTS (MIDEA)	<i>Lead supervisor</i> 2018
Chelsea Ellen Claire Owensby MASTER OF INTERACTION DESIGN AND ELECTRONIC ARTS (MIDEA)	<i>Co-supervisor</i> 2018

Academic Community

2025	Diversity Chair , Australian Conference on Human-Computer Interaction (OzCHI)	<i>Sydney, Australia</i>
2025	Workshops Co-Chair , International Symposium on Mixed and Augmented Reality (ISMAR)	<i>Daejeon, South Korea</i>
2024	Tutorials Co-Chair , International Symposium on Mixed and Augmented Reality (ISMAR)	<i>Seattle, USA</i>
2024	Workshops and Tutorials Co-Chair , UbiComp / ISWC 2024	<i>Melbourne, Australia</i>
2022	Publication Chair , ACM Interactive Surfaces and Spaces Conference (ISS)	<i>Wellington, New Zealand</i>
2020	Late Breaking Work Chair , Australian Conference on Human-Computer-Interaction (OzCHI)	<i>Sydney, Australia</i>
2019	Short Paper Chair , Australian Conference on Human-Computer-Interaction (OzCHI)	<i>Perth, Australia</i>
2020	Publication Chair , Media Architecture Biennale (MAB)	<i>Amsterdam, The Netherlands</i>
2018	Publication Chair , Media Architecture Biennale (MAB)	<i>Beijing, China</i>
2016	Publication Chair , Australian Conference on Human-Computer-Interaction (OzCHI)	<i>Launceston, Australia</i>
2016	Student Competition Chair , Media Architecture Biennale (MAB)	<i>Sydney, Australia</i>

Grants & Funding

PRINCIPAL INVESTIGATOR

2023	Provost's CAPEX and Contingency Fund Round 3 , Secured funding to purchase a portable mixed reality (MR) system, "Building Context-Based Prototyping Capacity through Mobile MR" with \$20,000 AUD.	<i>The University of Sydney</i>
2022	Provost's CAPEX and Contingency Fund Round 2 , Secured funding to purchase haptic gloves for virtual reality simulations, "Advancing Capability in Haptic Interactions for Virtual Reality and Robotics" with \$118,500 AUD.	<i>The University of Sydney</i>
2022	Provost's CAPEX and Contingency Fund Round 1 , Secured funding to purchase haptic equipment for virtual reality simulations, "Building capability in motion capture and tracking" with \$29,441 AUD.	<i>The University of Sydney</i>

CO-INVESTIGATOR

2025	Sydney Environment Institute's Collaborative Grant Scheme , Funding gained to support a multidisciplinary collaborative project focused on <i>engaging human and non-human actors in crisis resilience: AI-supported immersive technologies for inclusive decision-making</i> with \$40,000 AUD.	The University of Sydney
2024	Sydney Southeast Asia Centre Incubator Grant (2025 funding) , Funding gained to support a collaborative project with VNUHCM University of Science (Vietnam) focused on <i>advancing crisis resilience in southeast asia through immersive technologies and AI-supported community engagement</i> with \$19,000 AUD.	The University of Sydney
2023	Provost's CAPEX and Contingency Fund Round 3 , In collaboration with researchers from the Faculty of Engineering and the Faculty of Medicine and Health, we secured a large CAPEX grant to develop a digital health hub at Westmead Hospital in the Western Sydney Local Health District, <i>"EmTech Health Hub: building a digitally capable health workforce via innovating services and products"</i> with \$555,000 AUD.	The University of Sydney
2022	Australian Research Council Discovery Project Grant , Our team's Australian Research Council (ARC) Discovery Research proposal (DP220102019) titled <i>"Shared-space interactions between people and autonomous vehicles"</i> was amongst 19% that were funded with \$468,525 AUD.	The University of Sydney
2020	Strategic Education Grant (SEG) , Our grant submission was titled <i>"Sydney DDI Toolkit: Building Students Capabilities for Designing Digital Interfaces"</i> . The project focused on the creation of an interface design toolkit that would take the form of a book. We secured \$8,000 AUD of funds for this project.	The University of Sydney

INDIVIDUAL AWARDS

2019	Tsinghua University Travel Grant , Secured funding from my School to support my travel to visit Tsinghua University, along with support for PhD students from Tsinghua University to visit the University of Sydney.	Beijing, China and Sydney, Australia
2018	SIGCHI Travel Grant , Secured funding from Association for Computing Machinery (ACM) Special Interest Group on Computer-Human Interaction (SIGCHI) for travel to Singapore to attend the Ubiquitous Computing conference (Ubicomp) 2018.	Singapore

Visiting Research

Visiting Scholar (Sabbatical)

UCLIC (UCL INTERACTION CENTRE) AND GLOBAL DISABILITY INNOVATION HUB (GDI HUB)

[London, United Kingdom](#)

February 2023 - June 2023

- Visited Associate Professor Youngjun Cho to build collaboration.
- Co-supervised a masters capstone student working on a project investigating test taking/exams within virtual environments.
- Co-organised a workshop at CHI 2023, *"Towards an Inclusive and Accessible Metaverse"*.

Visiting Scholar

THE FUTURE LAB, TSINGHUA UNIVERSITY

[Beijing, China](#)

December 2019

- Supervised and worked with researchers and PhD students at The Future Lab on a project exploring the use of selfie-taking drones for engaging elderly citizens with their local communities.

Visiting Scholar

GSCT UBIQUITOUS VIRTUAL REALITY LABORATORY (UVR LAB), KAIST

[Daejeon, South Korea](#)

December 2018 - January 2019

- Worked with researchers and PhD students at the lab to understand the effect of room size and furniture on presence, narrative engagement, and usability during a space-adaptive augmented reality (AR) game.

Invited Talks

Keynote: The metaverse - Towards accessibility and inclusivity

THE THIRD WORKSHOP ON BUILDING AN INCLUSIVE AND ACCESSIBLE METAVERSE FOR ALL AT CHI 2025

[Yokohama, Japan](#)

January 2025

Inclusivity and Accessibility in the Metaverse

KOREA ADVANCED INSTITUTE OF SCIENCE AND TECHNOLOGY (KAIST) & SEOUL NATIONAL UNIVERSITY

[South Korea](#)

January 2025

Augmenting spaces for the people

MICROSOFT RESEARCH CAMBRIDGE

[Cambridge, United Kingdom](#)

May 2023

Designing Accessible Technologies in Future Cities

LANCASTER UNIVERSITY

[Lancaster, United Kingdom](#)

May 2023

Human anatomy, 3D modelling and Augmented reality

DEPARTMENT OF COMPUTER ENGINEERING, MOKPO NATIONAL UNIVERSITY

[Mokpo, South Korea](#)

October 2022

Designing Assembly Instruction Representations for Augmented Reality Displays

DESIGN LUNCHTIME RESEARCH SEMINAR, THE UNIVERSITY OF SYDNEY

[Sydney, Australia](#)

June 2021

Towards augmented, accessible cities

THE FUTURE LAB, TSINGHUA UNIVERSITY

[Beijing, China](#)

December 2020

Augmented Urban Interfaces

URBAN INTERFACES LAB SEMINAR, THE UNIVERSITY OF SYDNEY

[Sydney, Australia](#)

October 2020

China-Australia-Korea Joint workshop on Augmenting Cities

THE FUTURE LAB, TSINGHUA UNIVERSITY

[Beijing, China](#)

October 2019

Relevant Public Displays and Augmented Reality in Public Space

GSCT UBIQUITOUS VIRTUAL REALITY LABORATORY (UVR LAB), KAIST

[Daejeon, South Korea](#)

January 2018

Observing Public Displays in the Wild

COMPUTER-HUMAN ADAPTED INTERACTION (CHAI) SEMINAR

[Sydney, Australia](#)

March 2018

Augmented Reality for Community Engagement

COMMUNITY ENGAGEMENT SUMMIT 2016 AND 2017

[Sydney & Perth, Australia](#)

June 2016 & August 2017

Augmented Reality and Public Displays

AWE MEDIA WEARABLES/AR/VR MEETUP

[Sydney, Australia](#)

June 2017

Workshop Organisation

The Third Workshop on Building an Inclusive and Accessible Metaverse for All

CHI 2025

[Yokohama, Japan](#)

May 2025

Building a Metaverse for All: Opportunities and Challenges for Future Inclusive and Accessible Virtual Environments

CHI 2024

[Hawaii, USA](#)

May 2024

Towards an Inclusive and Accessible Metaverse

CHI 2023

[Hamburg, Germany](#)

April 2023

Designing Smart for Sustainable and Resilient Communities: The Role of Participatory Design in Addressing the UN Sustainable Development Goals

PARTICIPATORY DESIGN CONFERENCE (PDC) 2022

[Brisbane, Australia](#)

August 2022

Augmenting Cities with Immersive Technologies Workshop

MEDIA ARCHITECTURE BIENNALE (MAB) 2018 & INTERNATIONAL SYMPOSIUM ON MIXED AND AUGMENTED REALITY (ISMAR)

2019

[Beijing, China](#)

2018 and 2019

Designing smart for sustainable communities: Reflecting on the role of HCI for addressing the sustainable development goals

AUSTRALIAN CONFERENCE ON HUMAN-COMPUTER INTERACTION (OZCHI) 2019

[Fremantle, Australia](#)

December 2019